

Steam Game Review Sentiment Analysis

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Why it matters

Performing sentiment analysis on game reviews allows developers to gain insights into player opinions and emotions, identify areas for improvement, and inform game development and marketing strategies. By understanding player sentiment, developers can create more engaging experiences, improve player satisfaction and retention, and ultimately drive business success in the competitive gaming industry.

Forgemaster Steam Reviews Dataset

Sourced from Kaggle. Itself sourced by scraping
the Steam API in 2021.

Comprises:

6,976,390 unique users

15,437,471 reviews

8,183 games reviewed

Spread across 11 csv files (~5 gigs worth)

Methods:

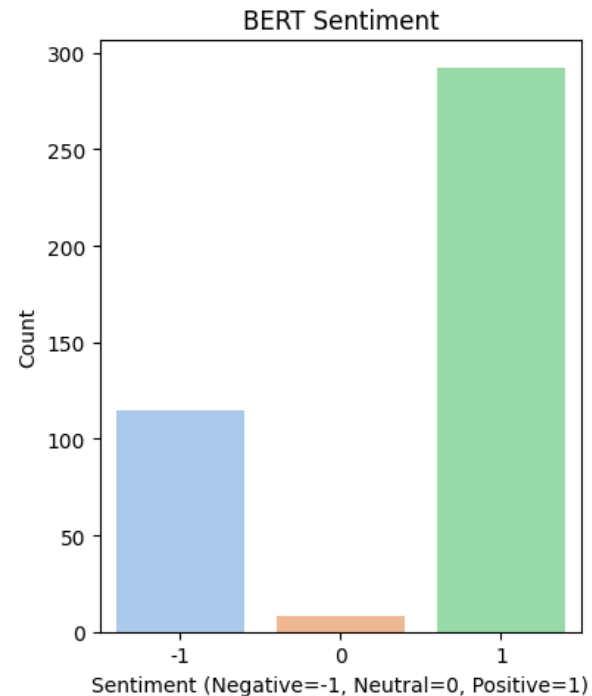
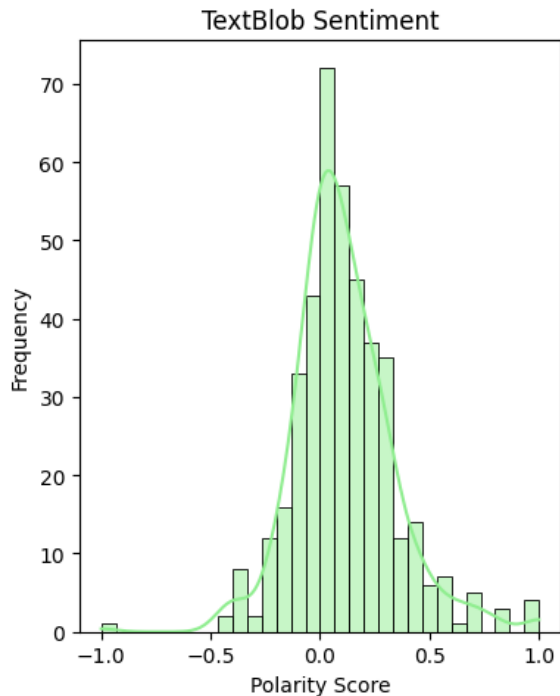
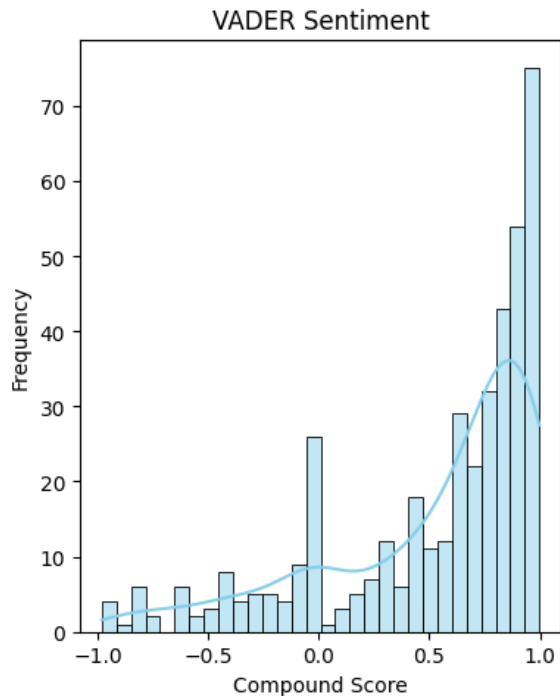
BERT Sentiment
TF-IDF
Logistic Regression
VADER
Text-Blob
K-Means Clustering
DBSCAN
Agglomerative Clustering
t-SNE

Preprocessing:

- Got rid of non-english reviews
- Made every word lowercase
- Removed stop words
- Removed whitespace
- Tokenized the text
- UMAP for dimension reduction

Results

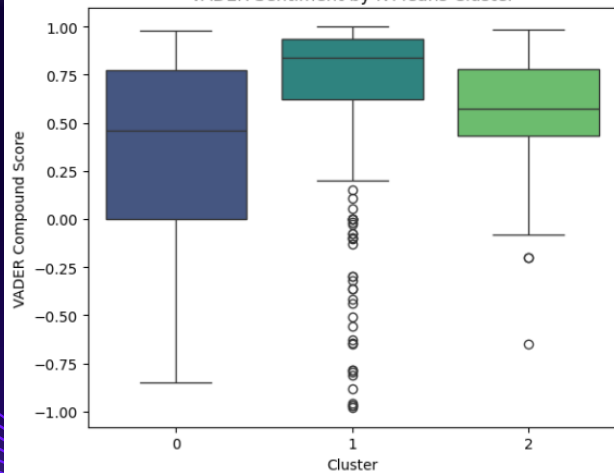
Distribution of Sentiment Scores Across Methods
(reviews-6805-9035.csv)



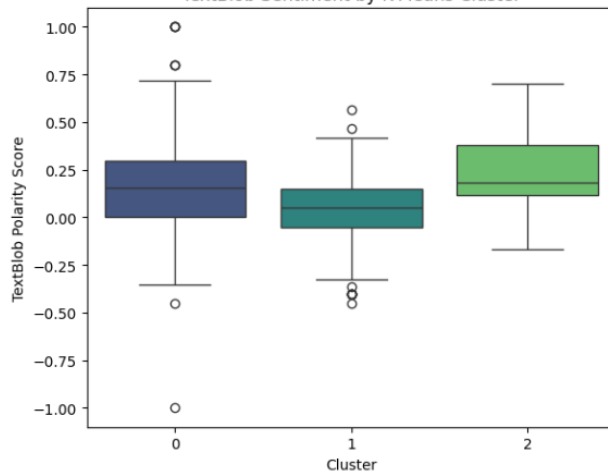
Results

Sentiment by K-Means Cluster Across Methods
(reviews-6805-9035.csv)

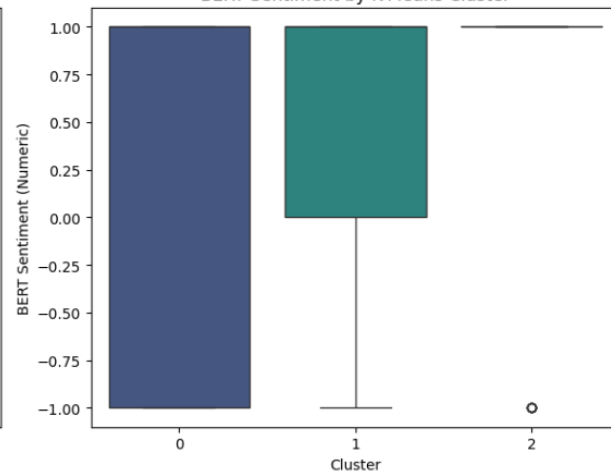
VADER Sentiment by K-Means Cluster



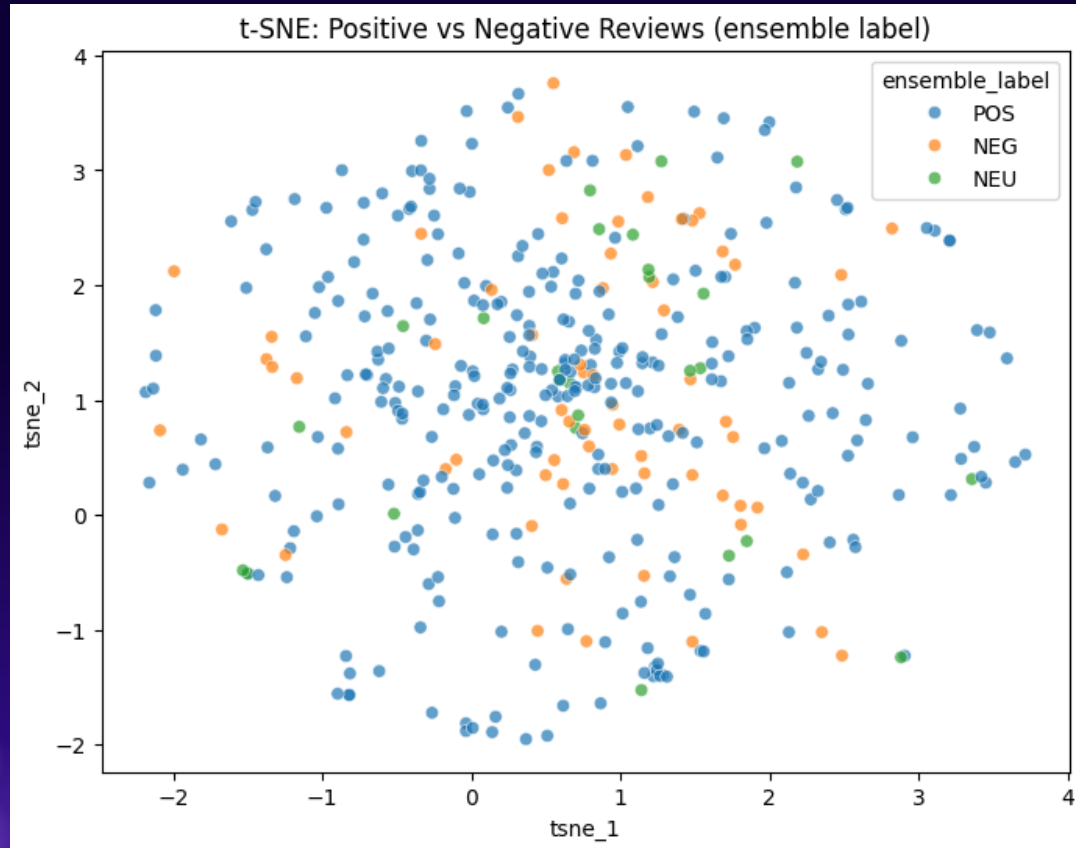
TextBlob Sentiment by K-Means Cluster



BERT Sentiment by K-Means Cluster



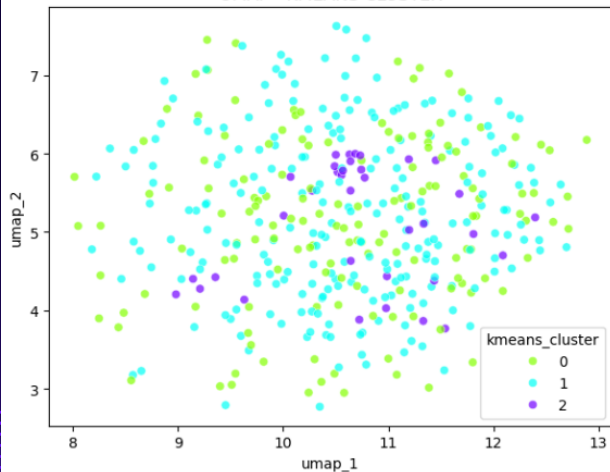
Results



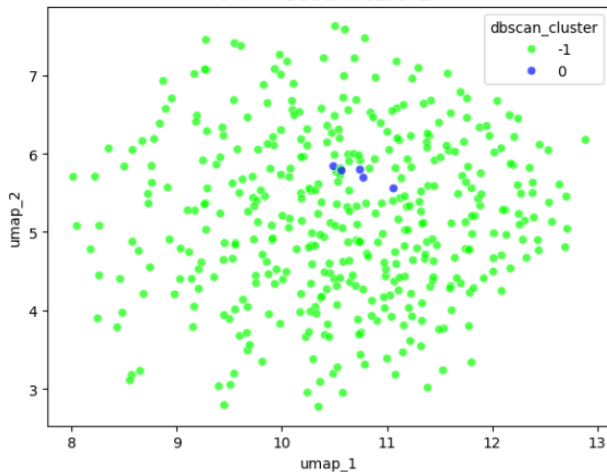
Results

UMAP Visualization Across Clustering Algorithms
(reviews-6805-9035.csv)

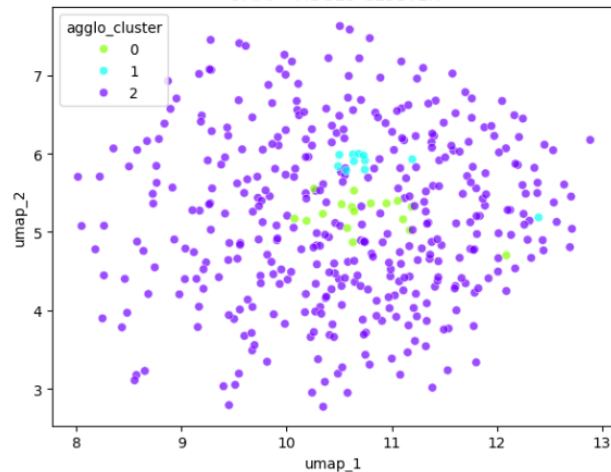
UMAP - KMEANS CLUSTER



UMAP - DBSCAN CLUSTER



UMAP - AGGLO CLUSTER



KMeans: Silhouette=0.014, Davies-Bouldin=7.347
DBSCAN: Not enough clusters (or only noise) for metrics.
Agglomerative: Silhouette=0.009, Davies-Bouldin=3.561

Results

TF-IDF + Logistic Regression Evaluation:

	precision	recall	f1-score	support
0	0.56	0.42	0.48	12
1	0.91	0.94	0.92	71
accuracy			0.87	83
macro avg	0.73	0.68	0.70	83
weighted avg	0.85	0.87	0.86	83

Challenges

One rather large issue we encountered was processing such a large (~5gb) dataset. We solved this by splitting the dataset into 11 smaller chunks.



In Conclusion

We used a combination of BERT Sentiment, TF-IDF, Logistic Regression, VADER, Text-Blob, K-Means Clustering, DBSCAN, Agglomerative Clustering, and t-SNE on steam reviews to perform sentiment analysis.

